NIRAV SHAH

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SUMMARY

Software engineer with 4+ years building high scale distributed systems on Linux and cloud in C++, Python and SQL. At Yahoo, re-architected core event processing to cut cloud cost 94% ($150K+/month), ran 50+ production services processing 5B+ events/day at a 99.9% SLA, and was promoted to SDE II within 2 years.

EXPERIENCE

Yahoo (Mountain View, CA)

Software Development Engineer II Mar 2026 - Present

- Re-architected 5 high-volume event pipelines from Pig on Dataproc into 3 partitioned BigQuery SQL jobs, cutting infrastructure cost 94% ($161K to $9K/month) and runtime 85% while improving scalability.
- Designed a near real-time analytics pipeline (Python, SQL and Airflow) ingesting 4M events/hour at 10-minute freshness for $1.2K/month, serving 100+ dashboard users across 10+ teams.
- Root-caused a production defect in event-processing logic inflating engagement 28% for Yahoo News, Homepage and Sports, tracing 884K phantom sessions and shipping the fix.
- Built, deployed and validated an Apache Beam (Dataflow) service change after an upstream source was retired, moving enrichment to the new source and restoring 85% of real-time attributes.
- Built a Claude AI-agent workflow automating schema changes, cutting each change from 2 weeks to 2 days.

Software Development Engineer I Feb 2024 - Feb 2026

- Led and mentored a team of 8 engineers migrating batch workloads from on-prem Oozie and Kubernetes to GCP Cloud Composer, redesigning 10+ workflows in Python with automated tests (pytest) and cutting peak runtime 20%.
- Maintained 50+ distributed streaming services (Kafka, Pub/Sub, Storm) on Linux at 99.9% SLA as on-call owner, diagnosing TCP timeouts and packet loss with `tcpdump` and `ss`, tuning socket and connection settings, cutting incident response time 50%.
- Debugged production failures on Linux hosts and containers for these services, tracing OOM kills, file-descriptor and disk exhaustion through system logs, `strace`, `top` and `lsof`, cutting repeat incidents 20%.
- Redesigned classification logic for an audience measurement system, cutting uncategorized pageviews 92%.

Software Development Engineer Intern May 2023 - Aug 2023

- Built a query benchmarking tool using a REST API to tag BigQuery jobs, with a UI to benchmark multiple date ranges storing runtime stats in Spanner, flagging 50 slow queries behind 7+ API endpoints.

PROJECTS

Simple FTP over UDP (C++, Linux): Built a reliable file transfer protocol with Go-Back-N ARQ, checksums, cumulative ACKs and retransmission; simulated packet loss and benchmarked delay vs window size and loss rate.

P2P File Sharing over TCP sockets (C++/Python): Designed a custom request/response application layer protocol with a multi-threaded index server handling concurrent peer joins, lookups and peer-to-peer file transfers.

Xinu OS Kernel (C, Linux): Implemented process scheduling, demand paging and reader-writer locks with priority inheritance inside the kernel; debugged with GDB on QEMU.

EDUCATION

North Carolina State University (Master of Computer Science) **GPA: 4.0/4.0** Aug 2022- Dec 2023
Pune Institute of Computer Technology (B.E. in Computer Engg) **GPA: 8.99/10** Aug 2016- May 2020

TECHNICAL SKILLS

Languages: C++, C, Python, SQL, Java, JavaScript
Systems & Networking: Linux, TCP/IP, UDP, socket programming, multithreading, strace, tcpdump, GDB
Infrastructure: Kubernetes, Docker, Terraform, CI/CD (Jenkins, GitHub Actions), Git, Bash, REST APIs
Data Systems: Kafka, Pub/Sub, BigQuery, Spark, Apache Beam (Dataflow), Airflow, Spanner